Lec-1

03/10/26 9:54 AM

Code

$x = 5$ ✓

$x = 5.5$ ✓ → Floating

$x = Ram$

Data TH

Java divides the into the following groups:

- ~~Arithmetic operators~~
- Assignment operators
- Comparison operators
- Logical operators
- Bitwise operators → × Bit-masking

Arithmetic Operators

Arithmetic operators are used to perform common mathematical operations.

Operator	Name	Description	Example	Try it
+	Addition	Adds together two values	x + y	Try it »
-	Subtraction	Subtracts one value from another	x - y	Try it »
*	Multiplication	Multiplies two values	x * y	Try it »
/	Division	Divides one value by another	x / y	Try it »
%	Modulus	Returns the division remainder	x % y	Try it »
++	Increment	Increases the value of a variable by 1	++x	Try it »
--	Decrement	Decreases the value of a variable by 1	--x	Try it »

ADVERTISEMENT

$64/7 = 9$

$64\%7 = 1$

$-3\%5 = -3$, 2

$-3+5 = 2$

$3\%5 = 3$

x++++

Java Assignment Operators

Assignment operators are used to assign values to variables.
In the example below, we use the **assignment** operator (=) to assign the value **10** to a variable called **x**:

Example

```
intx =10;
```

Try it Yourself »

The **addition assignment** operator (+=) adds a value to a variable:

Example

```
intx =10;x +=5;
```

Try it Yourself »

A list of all assignment operators:

Operator	Example	Same As	Try it
=	x = 5	x = 5	Try it »
+=	x += 3	x = x + 3	Try it »
-=	x -= 3	x = x - 3	Try it »
*=	x *= 3	x = x * 3	Try it »
/=	x /= 3	x = x / 3	Try it »
%=	x %= 3	x = x % 3	Try it »
&=	x &= 3	x = x & 3	Try it »
|=	x |= 3	x = x | 3	Try it »
^=	x ^= 3	x = x ^ 3	Try it »
>>=	x >>= 3	x = x >> 3	Try it »
<<=	x <<= 3	x = x << 3	Try it »

x

=+

$x = 5$ → Assignment

$x == 5$

+=

$x = x + 3$

$x += 3$

$x += 3$

$x =+ 3$

+=
-=
%=
*=
/=

$x += 3$
↳ $x = x + 3$

$x /= 2$
↳ $x = x/2$

$x -= 3$
↳ $x = x - 3$

$x == y$

$x >= y$

$x <= y$

$x > y$

$x < y$

Java Comparison Operators

Comparison operators are used to compare two values:

Operator	Name	Example	Try it
==	Equal to	x == y	Try it »
!=	Not equal	x != y	Try it »
>	Greater than	x > y	Try it »
<	Less than	x < y	Try it »
>=	Greater than or equal to	x >= y	Try it »
<=	Less than or equal to	x <= y	Try it »

Java Logical Operators

Logical operators are used to determine the logic between variables or values:

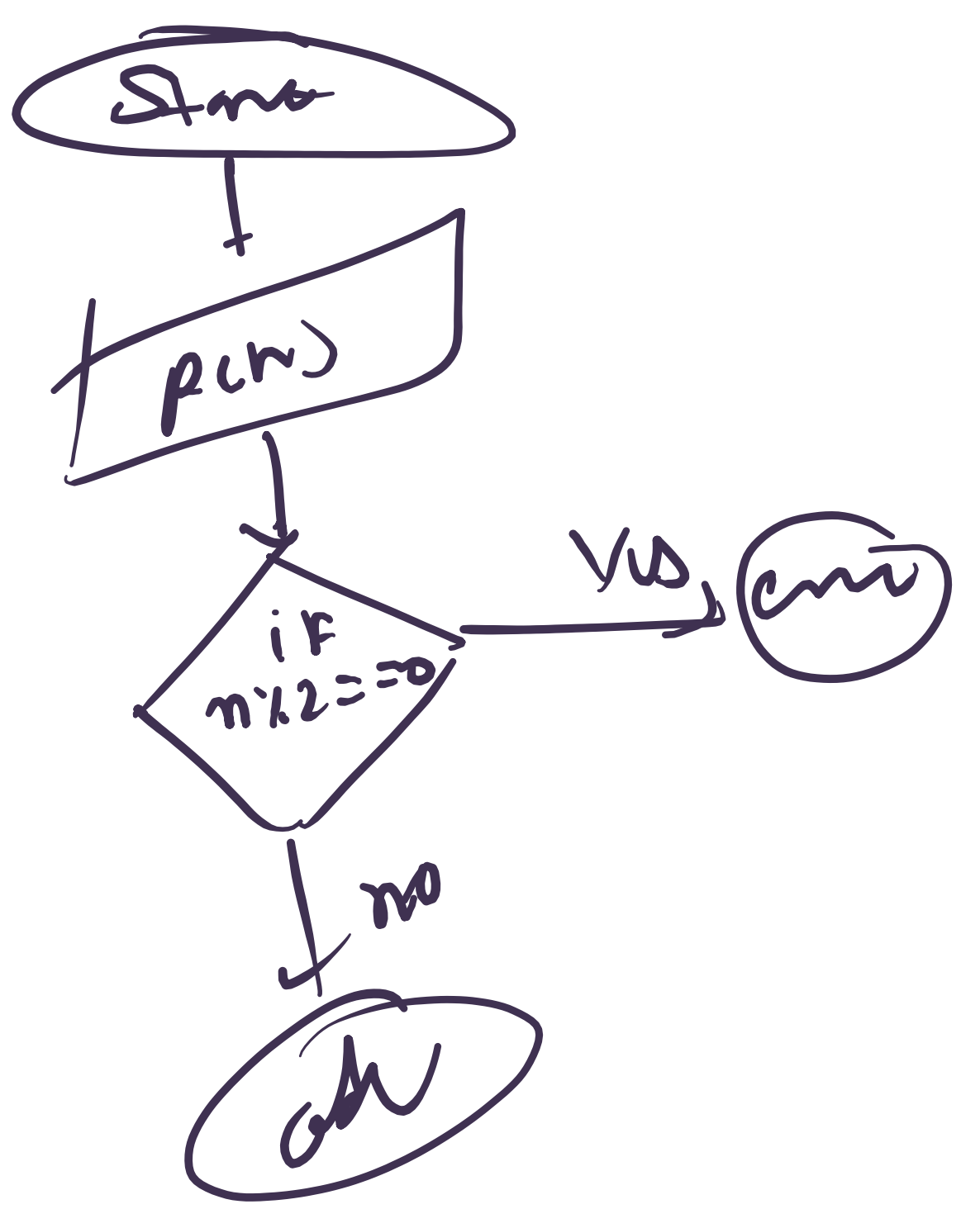
Operator	Name	Description	Example	Try it
&&	Logical and	Returns true if both statements are true	x < 5 && x < 10	Try it »
||	Logical or	Returns true if one of the statements is true	x < 5 || x < 4	Try it »
!	Logical not	Reverse the result, returns false if the result is true		

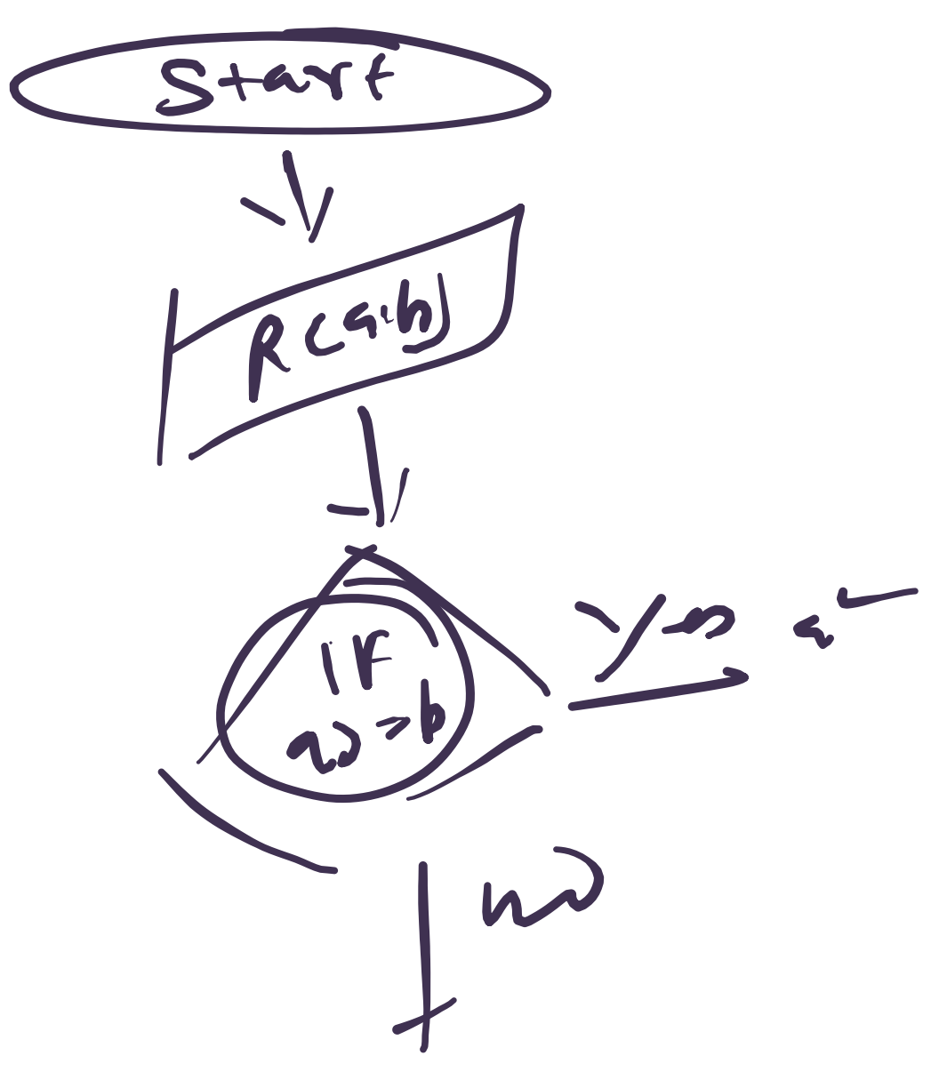
!(T) = F

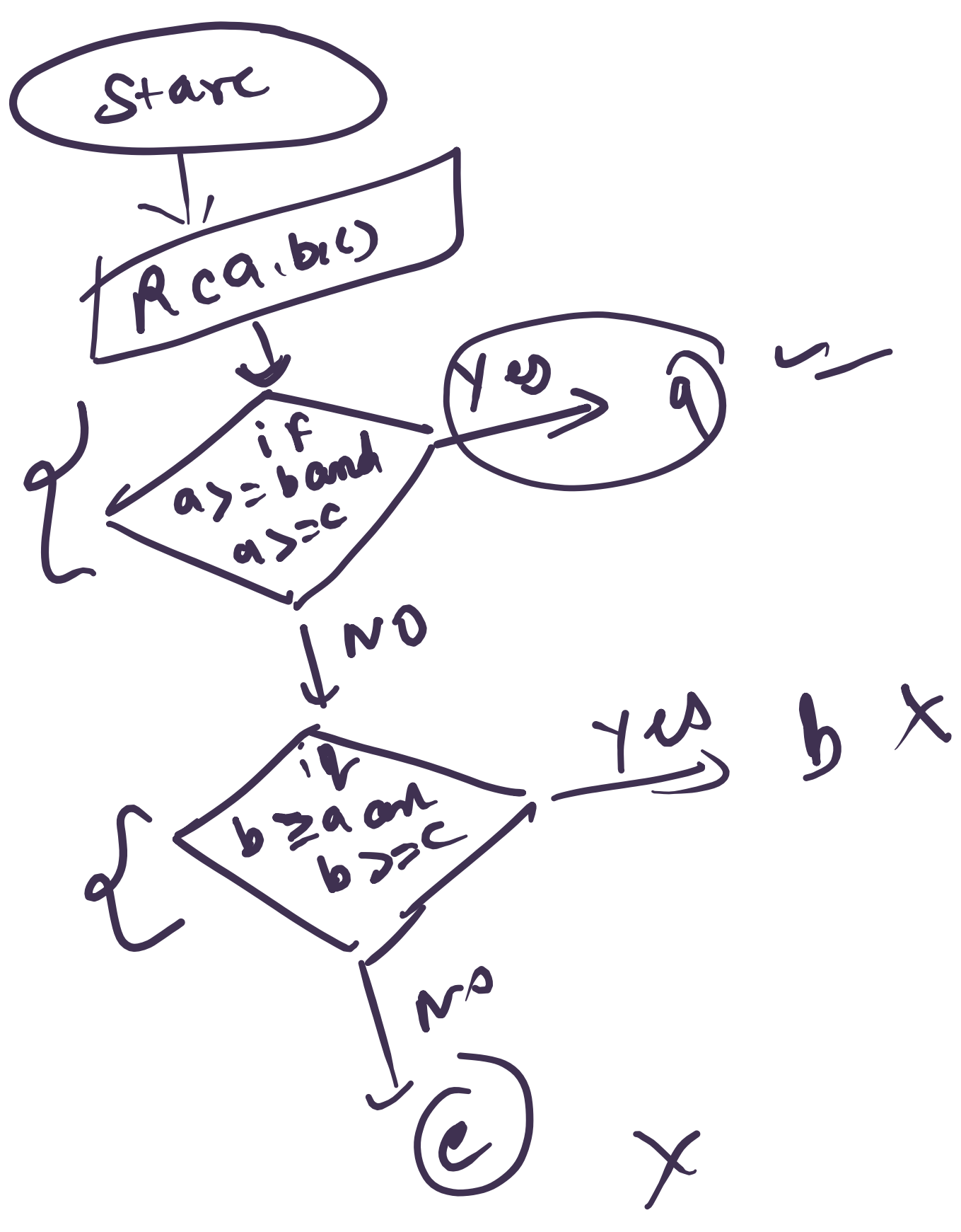
!(F) = T

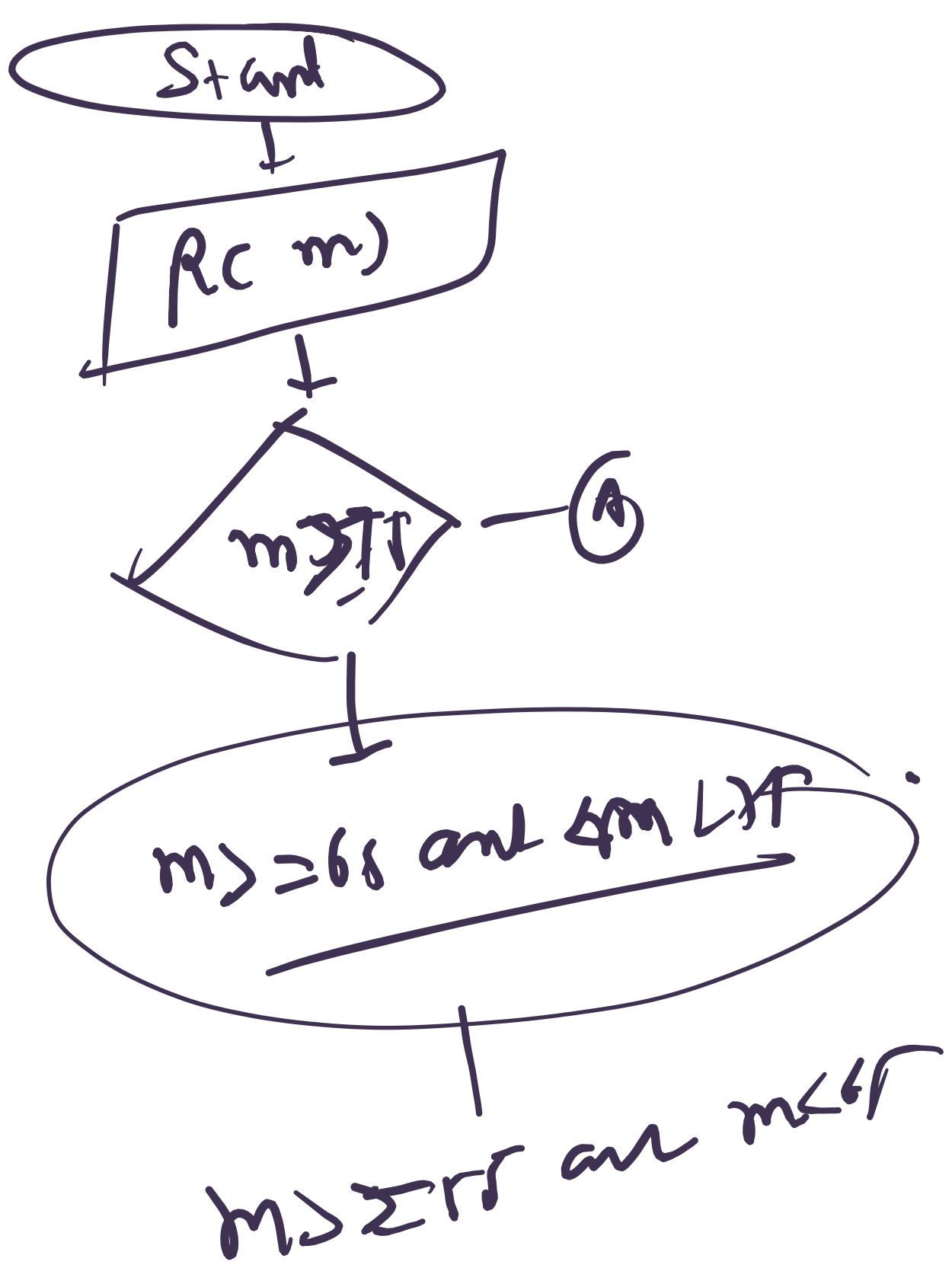
a	b	a&&b	a||b
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

$n\%2 == 0$









```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int x = 8;
//      System.out.println(x++);// --> x = x + 1;
//      System.out.println(x);

//      System.out.println(--x);// --> x = x - 1;
//      System.out.println(x);
        int a = x++ + --x + x-- + --x;
        System.out.println(a);

    }
```

$x = \cancel{8}\ \cancel{9}\ \cancel{8}\ \cancel{7}\ 6$

$a = 8 + 8 + 8 + 6$

$= 30$ ✓

```
int a = x++ - ++x - x-- - --x + ++x;
        System.out.println(a);
```

$x = \cancel{8}\ \cancel{9}\ \cancel{10}\ \cancel{9}\ \cancel{8}\ 9$

$\cancel{8} - 10 - 10 - \cancel{8} + 9$

$-20 + 9 = -11$

ⅰ $n \geq 100$ && $n \leq 200$ Bike

ⅱ $n \geq 282$ && $n \leq 473$ macbook

ⅲ $n \geq 50$ $n \leq 93$ cycle

ⅳ $n \geq 789$ $n \leq 989$ Kurkure

ⅴ Better luck next time

Bike → $n >= 100$ $n \leq 150$ ↳ Hero

Bike → KTM

macbook → $n > 282$ $n \leq 50$ ↳ m_1

macbook → m_2